

Computational Identification of Unknown Organic Contaminants and Molecules in Remote Environments via High-Resolution Mass Spectrometry

Raelyn Brooks

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1 Goal

Identification of unknown compounds by High-Resolution Mass Spectrometry (HRMS) in water samples from snow (precipitate) in the Himalayas/Tibet, with a particular focus on the long-range atmospheric transport of these organic contaminants to these remote areas. The identification will be made using computational tools and software, including molecular networking or cluster analysis.

2 Introduction to Environmental Chemistry and the Widespread Problem of Organic Contamination

2.1 The Global Problem: Widespread Dispersion of Anthropogenic Chemicals

The continuous release of anthropogenic chemicals, including herbicides, pharmaceuticals, industrial compounds, and plasticizers, has led to their pervasive presence in remote environments. These contaminants, often originating from human activities, are spread globally. Specific examples of concern include pesticides (e.g., atrazine, detected across sampling sites), industrial compounds, polycyclic aromatic hydrocarbons (PAHs) from fossil fuel combustion and vehicle traffic, and plasticizers like dimethyl phthalate. Anthropogenic markers such as vanillin also indicate human influence on remote environments. The widespread nature of these chemicals poses significant environmental and ecological risks, necessitating comprehensive monitoring and mitigation strategies.

2.2 Why Remote Peaks? Kilimanjaro and the Himalayas as "Cold Traps"

Remote high-altitude regions such as the Qinghai-Tibetan Plateau (QTP), Mount Everest, Yuzhu Mountain, Haba Mountain, and Kilimanjaro serve as crucial indicators of global pollution. These areas generally lack significant local industrial activity, meaning that any detected contaminants are strong evidence of long-range atmospheric transport. The cold temperatures characteristic of

these mountain environments slow the degradation rates of persistent organic pollutants (POPs), allowing them to accumulate in these remote ecosystems. This "cold-trapping" effect concentrates contaminants, making these regions particularly vulnerable to persistent global pollution.

2.3 Study Goal: Using HRMS to Move Beyond "Target Lists" and Identify the "Known Unknowns" in the Environment

Traditional targeted analyses often focus on a predefined list of known contaminants. However, this approach can miss a vast array of "known unknowns" and entirely unknown compounds present in environmental samples. This project aims to employ non-targeted analysis and suspect screening using HRMS to comprehensively identify organic contaminants in high-altitude snow and water samples. This methodology allows for the detection and characterization of a broader spectrum of pollutants, thereby enriching the understanding of contaminant fate and transport.

3 Atmospheric Chemistry & Transport Mechanisms

3.1 Atmospheric Entry: Primary Sources and Chemical Makeup

Organic contaminants enter the atmosphere through various primary sources, including industrial volatility, agricultural pesticide drift, fossil fuel combustion, burning of biomass, and vehicle emissions. Once airborne, these compounds can undergo various chemical transformations and interactions within the troposphere, influencing their persistence, reactivity, and transport potential.

3.2 The Grasshopper Effect (Multi-Hop Transport)

The "grasshopper effect" describes the cyclical process of volatilization, atmospheric transport, condensation, and deposition of semi-volatile organic compounds (SVOCs) from warmer to colder regions. This mechanism explains the accumulation of POPs in high-latitude and high-altitude "cold traps". This iterative process allows contaminants to travel vast distances from their emission sources to remote environments.

3.3 Deposition Physics: Snow as an Efficient "Scrubber" of Organic Pollutants

Snowfall serves as a highly efficient mechanism for the atmospheric scavenging and deposition of organic pollutants. Compared to rain, snow is often considered a more effective "scrubber" of organic chemicals due to its larger surface area and the process of cold condensation. As snow melts, these accumulated contaminants are re-released into the environment, potentially impacting local hydrology and ecosystems. The study of snow samples provides a valuable proxy for atmospheric deposition, reflecting long-range transport patterns.

3.4 Regional Pathways: Indian Summer Monsoon and "Atmospheric Brown Clouds"

In regions like the Tibetan Plateau, specific meteorological phenomena play a critical role in contaminant transport. The Indian Summer Monsoon (ISM) and "Atmospheric Brown Clouds" (ABCs), laden with pollutants from the densely populated Indo-Gangetic Plain, act as significant carriers of pollution. These systems facilitate the upward and northward transport of a wide range of organic contaminants, depositing them onto the Himalayan and Tibetan ranges. Understanding these regional atmospheric pathways is essential for tracing the origins and predicting the deposition patterns of contaminants in these high-altitude ecosystems.

4 Analytical Methodology: The HRMS Workflow for Non-Target Identification

4.1 How and why NTA is useful to environmental chemists

- **Broader Exposure Characterization:** NTA allows for the characterization of human exposures in various environmental media (air, water, dust, soil) and human samples, moving beyond the limited scope of traditionally assessed pollutants.
- **Identification of Emerging Contaminants:** It helps identify potential chemical stressors, including transformation and degradation products, and prioritizes emerging contaminants for further safety assessment, crucial for understanding complex environmental chemistry.
- **Support for Rapid Response:** NTA can be deployed for rapid response in environmental emergencies, such as wildfires or unintended chemical releases.
- **Comprehensive Environmental View:** By not being limited to predefined target lists, NTA provides a more comprehensive understanding of environmental samples, leading to the discovery of previously unmonitored or unknown pollutants.

4.2 Current challenges and limitations for NTA in environmental chemistry

- **Lack of Standardization:** Practices in hardware, software, databases, algorithms, and reporting strategies vary across institutions, laboratories, and investigators, hindering transparency and reproducibility, particularly for regulatory applications.
- **Data Processing and Interpretation:** The high volume and complexity of HRMS data pose challenges in processing, annotating, and prioritizing chemical candidates, despite the development of tools like the US EPA's INTERPRET NTA.
- **Identification Certainty:** Assigning structures to detected features, especially novel or unknown ones, remains challenging, often leaving many unidentified features in datasets.

- **Database Limitations:** The accuracy of suspect screening relies heavily on the comprehensiveness and quality of chemical suspect databases and experimental MS/MS libraries, with *in silico* analysis having its own limitations.
- **Chemical Space Coverage:** While NTA expands the detectable chemical space, certain compound classes (e.g., very large molecules >1500 Da and metals) still require specialized analytical techniques beyond standard LC/GC-HRMS.

4.3 Example studies from the literature that applied NTA to environmental samples

A study used LC-HRMS for suspect screening and NTA of fresh snowfall samples in the central United States. It tentatively identified 91 compounds, confirming and semi-quantifying 17, mostly anthropogenic, including herbicides, insect repellents, and plasticizers. Prominent compounds found in all samples included N-cyclohexylformamide (tire leachate contaminant), DEET (insect repellent), and dimethyl phthalate (plasticizer)

4.4 Detection Precision: Why HRMS (Orbitrap or Q-TOF) is Required

High-Resolution Mass Spectrometry (HRMS), specifically instruments like Orbitrap or Quadrupole Time-of-Flight (Q-TOF), is crucial for non-target analysis due to its ability to provide "Exact Mass" measurements with very low mass error (typically <5 ppm). This high precision allows for the accurate determination of elemental compositions, which is the first critical step in identifying unknown compounds, differentiating them from isobaric interferences, and enhancing the confidence in structural elucidation.

4.5 From Samples to Data: Solid-Phase Extraction (SPE)

After collection, melted snow and water samples must undergo preparation to concentrate trace organic contaminants. Solid-Phase Extraction (SPE) is a widely used and effective method for this purpose. SPE selectively extracts and concentrates organic compounds from large volumes of aqueous samples, enhancing their detectability by HRMS while removing interfering matrix components.

4.6 Computational Identification and Molecular Networking: Moving Beyond Libraries

This powerful bioinformatics approach groups unknown molecules by their structural similarity based on their MS/MS fragmentation patterns. Molecular networking helps visualize relationships between compounds, identify transformation products, and prioritize features for further investigation, even if they are not in existing spectral libraries.

4.7 In Silico Tools:

For molecules not found in any existing database, advanced in silico tools such as SIRIUS, MetFrag, and CSI:FingerID are indispensable. These tools predict molecular formulas and potential structures by analyzing exact mass, isotopic patterns, and fragmentation spectra, significantly aiding in the structural elucidation of truly unknown environmental contaminants.

References

- [1] Fan, W. Suspect Screening and Semi-Quantification of Organic Contaminants in Snow and Water Samples on the Qinghai–Tibet Plateau and High Mountain Regions. Undergraduate Thesis, The College of Wooster, Wooster, OH, 2025; Unpublished thesis.
- [2] Kind, T.; Tsugawa, H.; Cajka, T.; Ma, Y.; Lai, Z.; Mehta, S. S.; Wohlgemuth, G.; Barupal, D. K.; Showalter, M. R.; Arita, M.; Fiehn, O. Identification of small molecules using accurate mass MS/MS search. *Mass Spectrometry Reviews* **2018**, *37*, 513–532.
- [3] Schymanski, E. L.; Jeon, J.; Gulde, R.; Fenner, K.; Ruff, M.; Singer, H. P.; Hollender, J. Identifying Small Molecules via High Resolution Mass Spectrometry: Communicating Confidence. *Environmental Science & Technology* **2014**, *48*, 2097–2098.
- [4] Ye, W.; Li, J.; Cai, X. Mfgnn: Multi-Scale Feature-Attentive Graph Neural Networks for Molecular Property Prediction. *Journal of Computational Chemistry* **2025**, *46*, e70011.
- [5] Manz, F. A.-B. J. M. F. Y. L. H. A. K. J. M. C. N. S. R. P. K. D. P. B. J. G. P. K. J. P. C. . Y. J. A., K. E. Non-targeted analysis (NTA) and suspect screening analysis (SSA): A review of examining the chemical exposome. *Journal of Exposure Science Environmental Epidemiology* **2023**, *33*(4), 524–536.
- [6] Nino-Suastegui, S.; Painter, E.; Sprankle, J. W.; Morrison, J. J.; Faust, J. A.; Gray, R. Non-targeted analysis and suspect screening of organic contaminants in temperate snowfall using liquid chromatography high-resolution mass spectrometry. *Environmental Research* **2025**, *266*, 120494.
- [7] Chao, A.; Minucci, J. M.; Ferland, T. M.; Carr, E. T.; Janesch, G.; Rizwan, S.; Whitehead, H. D.; Cathey, T.; Pu, S.; Brunelle, L. D.; Batt, A. L.; Sobus, J. R.; Williams, A. J. Prioritizing Chemical Candidates from Non-targeted Analysis Using Metadata, Spectral Similarity, and Hazard Scoring within INTERPRET NTA. *Analytical Chemistry* **2025**, *97*, 15904–15912, PMID: 40670324.
- [8] Fisher, P. K. N.-S. e. a., C.M. Approaches for assessing performance of high-resolution mass spectrometry–based non-targeted analysis methods. *Anal Bioanal Chem* **2022**, *414*, 6455–6471.

- [9] Schmid, D. P. L.-F. N. M. W. A. T. A. A. J. H. T. e. a., Robin Ion Identity Molecular Networking for Mass Spectrometry-based Metabolomics in the GNPS Environment. *Nature Communications* **2021**, *12*.
- [10] Ernst, M.; Kang, K. B.; Caraballo-Rodríguez, A. M.; Nothias, L.-F.; Wandy, J.; Chen, C.; Wang, M.; Rogers, S.; Medema, M. H.; Dorrestein, P. C.; van der Hooft, J. J. MolNetEnhancer: Enhanced Molecular Networks by Integrating Metabolome Mining and Annotation Tools. *Metabolites* **2019**, *9*.
- [11] Olivon, F.; Grelier, G.; Roussi, F.; Litaudon, M.; Touboul, D. MZmine 2 Data-Preprocessing To Enhance Molecular Networking Reliability. *Analytical Chemistry* **2017**, *89*, 7836–7840, PMID: 28644610.
- [12] Zhu, W.; Zhang, Y.; Zhao, D.; Xu, J.; Wang, L. HiGNN: A Hierarchical Informative Graph Neural Network for Molecular Property Prediction Equipped with Feature-Wise Attention. *Journal of Chemical Information and Modeling* **2023**, *63*, 43–55, PMID: 36519623.
- [13] Herbert, B.; Villa, S.; Halsall, C. Chemical interactions with snow: Understanding the behavior and fate of semi-volatile organic compounds in snow. *Ecotoxicology and Environmental Safety* **2006**, *63*, 3–16, The Role of High Mountains in the Global Transport of Persistent Organic Pollutants.
- [14] Daly, G. L.; Wania, F. Organic Contaminants in Mountains. *Environmental Science & Technology* **2005**, *39*, 385–398, PMID: 15707037.
- [15] Blais, J. M., Rosen, M. R., Smol, J. P., Eds. *Environmental Contaminants: Using Natural Archives to Track Sources and Long-Term Trends of Pollution*; Developments in Paleoenvironmental Research; Springer Dordrecht: Dordrecht, 2015; Vol. 18.
- [16] Xu, Y.; Zhang, G.; Li, J.; Chakraborty, P.; Li, H.; Liu, X. Long-range atmospheric transport of persistent organochlorinated compounds from south and mainland south-eastern Asia to a remote mountain site in south-western China. *Journal of environmental monitoring : JEM* **2011**, *13*, 3119–27.
- [17] Kiss, A.; Fries, E. Occurrence of benzotriazoles in the rivers Main, Hengstbach, and Hegbach (Germany). *Environmental Science and Pollution Research* **2009**, *16*, 702–710.
- [18] Gray, R.; Painter, E.; Sprankle, J. W.; Crawford, A. D.; Morrison, J. J.; Frazier, M.; Faust, J. A. Suspect screening for pesticides in rain and snow using liquid chromatography high-resolution mass spectrometry. *Atmospheric Environment* **2022**, *291*, 119389.